

# COMPACT SUPPORT OF AN ISENTROPIC IMPLOSION A CONDITIONAL BLUEPRINT

ABSTRACT. A compactly supported physical-vacuum realisation of a radial isentropic self-similar implosion is reduced to a mixed one-dimensional free-boundary problem. The interior coincidence with the profile on the sonic cone is proved. Existence of the vacuum collar is conditional on the energy pairings of Coutand–Shkoller, *Comm. Pure Appl. Math.* 64 (2011), Lemma 3.1 and Theorem 8.1, transferred to a chart with one lock and one vacuum end. This is not a resolution of the Clay Navier–Stokes problem.

## 1. SETTING

Radial isentropic Euler,  $d \in \{2, 3\}$ ,  $\gamma > 1$ . Profile: Merle–Raphaël–Rodnianski–Szeftel [4] or Buckmaster–Cao–Labora–Gómez–Serrano [1]. Sonic ray  $Z = Z_s$ ,  $r_s(t) = Z_s(-t)^\lambda$ . Collar  $[r_{\text{br}}, R_0]$  with physical vacuum at  $R_0$ . Lock set  $\mathcal{S}$ :

$$J_+(t, a_*) = \sigma(t) \quad \text{on } [t_0, 0), \quad \sigma(t) = \bar{J}_+(t, \bar{\eta}(t, \bar{a}_*)).$$

## 2. STRETCHED BRIDGE

$Z_0$  first, then  $L$ . No circular dependence.  $r_{\text{in}} = Z_0(-t_0)^\lambda$  so that

$$\max_{k=1,2,3} |\partial_r^k(u, c)|_{r_{\text{in}}} < \text{Lip}_*/8.$$

Then

$$L > \frac{35}{16} \frac{|q_0^{\text{br}} - q_0^{\text{in}}|}{\text{Lip}_*/2}.$$

$D_t J_+ = -c \partial_r J_+ - c \frac{d-1}{r} u$ ,  $\det = c^6 > 0$ . Degree-7 Hermite,  $C_* = 35/16$ .  $Z_\Gamma(t_0) > Z_s$ ,  $\mu(t_0) > 0$ .

$$\text{Lip}_* = \min(1/|t_0|, \mu(t_0)(-t_0)^{-\lambda}/4).$$

## 3. VARIATIONAL FORMULATION

$V_* = \{\varphi \in H^1(I) : \varphi(a_*) = 0\}$ . Not  $V_{00}$ . The edge is free:  $v(1) = \dot{R}$ . Cutoff  $\omega \equiv 0$  near  $a_*$ ,  $\omega \equiv 1$  near vacuum. Lift  $v_L$  supported in  $\{\omega = 0\}$ . Option B basis  $\phi_k(y) = (y - a_*)^k$ ,  $k \geq 1$ . The bound  $C(M_0, N)$  is only for the finite ODE. Uniform energy uses the continuous weak form on  $V_*$ .

## 4. ENERGY ESTIMATES

Split  $\mathcal{E}^{(N)} = \mathcal{E}_{\text{in}}^{(N)} + \mathcal{E}_{\text{vac}}^{(N)}$ . Fluxes at  $a_*$  die by  $\varphi(a_*) = 0$ . Fluxes at  $y = 1$  die by  $p(1) = \rho_0(1) = 0$ .

For  $\gamma \geq 2$  ( $a_0 = 0$ ):

$$(1) \quad \begin{aligned} & \frac{d}{dt} (\mathcal{E}_{\text{in}}^{(N)} + \mathcal{E}_{\text{vac}}^{(N)}) + \frac{7}{8} \kappa \int_I \omega \rho_0^2 |\partial_t^4 \partial_y w^{(N)}|^2 \\ & \leq C(M_0) \mathcal{E}^{(N)} + C \|\sigma\|_{W^{4,\infty}}^2. \end{aligned}$$

For  $\gamma \in (1, 2)$  add the Theorem 8.1 sum,  $d = 1 - \gamma$ ,  $a_0$  from  $1 < 1 + 1/(\gamma - 1) - a_0 \leq 2$ :

$$\sum_{a=0}^{a_0} \left\| \sqrt{d}^{1+1/(\gamma-1)-a} \partial_t^{4+a_0-a} \partial_y w^{(N)} \right\|_{L^2(\{\omega=1\})}^2.$$

Explicitly:

- $\gamma = 5/3$ ,  $a_0 = 1$ :  $\|d^{5/4} \partial_t^5 \partial_y w\|_{L^2}^2 + \|d^{3/4} \partial_t^4 \partial_y w\|_{L^2}^2$ .
- $\gamma = 7/5$ ,  $a_0 = 2$ :  $\|d^{7/4} \partial_t^6 \partial_y w\|_{L^2}^2 + \|d^{5/4} \partial_t^5 \partial_y w\|_{L^2}^2 + \|d^{3/4} \partial_t^4 \partial_y w\|_{L^2}^2$ .
- $\gamma = 2$ ,  $a_0 = 0$ : model energy only.

The line  $d^{1/4} \partial_t^5 \partial_y w$  is not Theorem 8.1.

$\mathcal{C}_4$  against  $\partial_t^4 w$ :  $j = 1$  by Lemma 3.1 on  $\partial_t^3 p \in H_0^1$ ;  $j = 2, 3$  by  $\mathcal{E}_{\text{vac}}$ ;  $j = 4$  by  $\|\partial_y p / \rho_0\|_\infty$  and Theorem 8.1. Young into the right-hand side of (1) plus  $\frac{1}{8} \kappa \int \omega \rho_0^2 |\partial_t^4 \partial_y w|^2$ .

Citation: [2], Lemma 3.1 and Theorem 8.1. Not Theorem 1.1.

$$T^* = \min \left( |t_0|, C(M_0)^{-1} \ln \left( 1 + \frac{C(M_0)M_0}{C(M_0)M_0 + C\|\sigma\|_{W^{4,\infty}}^2} \right) \right).$$

## 5. CONTINUATION AND THE THEOREM

Mesh  $t_k = t_0 + kT^*$ ,  $K = \lceil |t_0|/T^* \rceil$ .  $N \rightarrow \infty$ ,  $\kappa \rightarrow 0$ : Aubin–Lions on  $\{\omega = 1\}$ , Rellich on  $\{\omega = 0\}$ . Lock exact on  $V_*$ , limit in  $\mathcal{S}$ . Cone  $\mathcal{K} = \{(t, r) : r \leq Z_s(-t)^\lambda\}$ . Friedrichs system in  $U = (\frac{2}{\gamma-1}c, u)$ ,  $A_0 = I$ , source  $F = (-\frac{d-1}{r}cu, 0)$  [3]. On the profile,  $\partial\mathcal{K}$  is a sonic characteristic:  $\dot{r}_s = \bar{u} - \bar{c}$ . Then  $\delta U = 0$  on  $\mathcal{K}$ .

**Hypothesis 1 (H).** *The mixed  $\kappa$ -problem on  $V_*$  admits approximants in  $\mathcal{S}$  on  $[t_0, 0)$ , uniformly bounded in  $\mathcal{E}_{\text{collar}}$ , with  $\kappa \rightarrow 0$  remaining in  $\mathcal{S}$  and keeping physical vacuum.*

**Theorem 1 (conditional).** *Assume Hypothesis H and the pairings of [2], Lemma 3.1 and Theorem 8.1, on  $\{\omega = 1\}$ . There is a compactly supported physical-vacuum solution on  $[t_0, 0)$  with  $(\rho, u) = (\bar{\rho}, \bar{u})$  on  $\{Z \leq Z_s\}$ ,  $\rho = 0$  for  $r \geq R(t)$ , and  $R(t) > r_s(t)$ .*

**Remark 1.** Not Clay.

## REFERENCES

1. Tristan Buckmaster, Gonzalo Cao-Labora, and Javier Gómez-Serrano, *Smooth imploding solutions for 3D compressible fluids*, Forum Math. Pi **13** (2025), e6.
2. Daniel Coutand and Steve Shkoller, *Well-posedness in smooth function spaces for the moving-boundary 1-D compressible Euler equations in physical vacuum*, Comm. Pure Appl. Math. **64** (2011), no. 3, 328–366, Lemma 3.1 and Theorem 8.1; not Theorem 1.1.
3. Tosio Kato, *Quasi-linear equations of evolution, with applications to partial differential equations*, Spectral theory and differential equations, Lecture Notes in Math., vol. 448, Springer, Berlin, 1975, pp. 25–70.
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